Individual Lab 1 — Java Solutions

Problem 1: Bicycle Rental

Save this program as **BicycleRental.java**. It validates the entered times and calculates the hourly charge using the rate for each hour of the rental.

```
import java.util.Scanner;

public class BicycleRental {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        System.out.print("Enter starting time (0-23): ");
        int start = input.nextInt();

        System.out.print("Enter ending time (1-24): ");
        int end = input.nextInt();

        if (start < 0 || start > 23 || end < 1 || end > 24 || start >= end) {
            System.out.println("Invalid time. The ending time must be after the starting time, within one day.");
            input.close();
            return;
        }

        int total = 0;

        for (int hour = start; hour < end; hour++) {
            if (hour < 7 || hour >= 21) {
                total += 500;
            } else if (hour < 14 || hour >= 19) {
                total += 1000;
            } else {
                total += 1500;
            }
        }

        System.out.println("Rental duration: " + (end - start) + " hour(s)");
        System.out.println("Total rental cost: " + total + " RWF");

        input.close();
    }
}
```

Problem 2: Mushroom Identifier

Save this program as **MushroomIdentifier.java**. It asks three yes/no questions. The supplied facts do not uniquely distinguish every mushroom, so the program reports possible matches when necessary.

```
import java.util.Scanner;

public class MushroomIdentifier {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        System.out.println("Answer each question with yes or no.");
        boolean gills = ask(input, "Does the mushroom have gills?");
        boolean forest = ask(input, "Does the mushroom grow in a forest?");
        boolean ring = ask(input, "Does the mushroom have a ring?");

        if (!gills) {
            System.out.println("The mushroom is Cep de bordeaux.");
        } else if (forest && !ring) {
            System.out.println("The mushroom may be Girolle or Pied bleu.");
        } else if (forest && ring) {
            System.out.println("The mushroom is likely Amanite tue-mouche.");
        } else if (!forest && ring) {
            System.out.println("The mushroom may be Agaric jaunissant or Coprin chevelu.");
        } else {
            System.out.println("The mushroom cannot be identified uniquely from these answers.");
        }

        System.out.println("Note: The given facts do not distinguish Agaric jaunissant from Coprin chevelu.");
        input.close();
    }

    private static boolean ask(Scanner input, String question) {
        while (true) {
            System.out.print(question + " ");
            String answer = input.nextLine().trim().toLowerCase();

            if (answer.equals("yes") || answer.equals("y")) {
                return true;
            }

            if (answer.equals("no") || answer.equals("n")) {
                return false;
            }

            System.out.println("Please answer yes or no.");
        }
    }
}
```